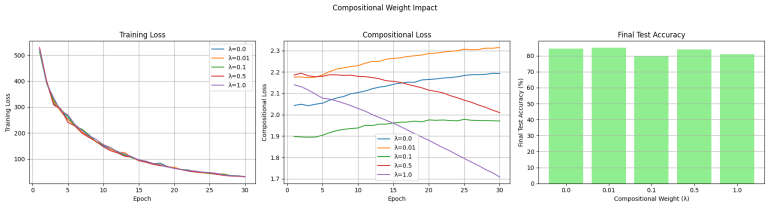

Figure 2: Impact of compositional weight λ on model performance. **Left**: Training loss over epochs for different λ. Higher λ values slightly increase training loss. **Middle**: Compositional loss decreases with higher λ, indicating the regularization term effectively enforces compositionality. **Right**: Final test accuracy does not improve with higher λ and may decrease, suggesting a trade-off between compositional regularization and the primary learning objective.

Comment: This is a stretch because the paper has not rigorously shown that lower compositional loss leads to more compositionality.

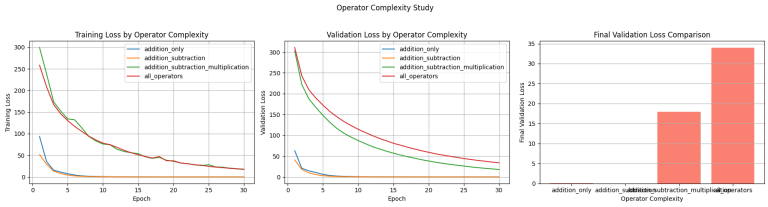

Figure 3: Model performance on expressions with varying operator complexity. **Left**: Training loss increases with operator complexity, indicating the models struggle to fit more complex data. **Middle**: Validation loss is higher for complex expressions, reflecting poor generalization. **Right**: Final validation accuracy decreases significantly as operator complexity increases, underscoring inherent limitations in handling complex compositional structures with compositional regularization alone.

Comment: This should be "Final validation loss increases" to match the figure, although the meaning remains roughly the same.

5 Conclusion

In this work, we introduced a compositional regularization term with the intention of enhancing compositional generalization in neural networks. Our experiments on synthetic arithmetic expression datasets revealed that compositional regularization did not lead to the expected improvements in generalization performance. In some cases, it even hindered the learning process. Additionally, we found that increasing the complexity of arithmetic expressions exacerbates the models' generalization difficulties, highlighting inherent limitations.

These findings highlight the challenges of enforcing compositional structures in neural networks through regularization. Possible reasons for the lack of improvement include conflicts between the regularization term and the primary learning objective, which may cause the network to prioritize minimizing the compositional loss over fitting the data. Additionally, the measure of compositionality used in the regularization term may not align with the aspects of compositionality that are critical for generalization. The synthetic dataset may also not adequately capture the complexities of compositional generalization in real-world tasks, and increased operator complexity introduces additional challenges that compositional regularization alone cannot overcome.

For future work, we suggest exploring alternative regularization strategies, refining the definition of compositionality in the context of neural networks, and testing on more complex datasets. Investigating models that can inherently handle higher operator complexity, such as those with recursive or hierarchical structures, may also be beneficial. Our findings underscore the importance of rigorously evaluating proposed methods and openly reporting negative or inconclusive results to advance our understanding of the challenges in deep learning.

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